2020 International Conference on Emerging Trends in Communication, Control and Computing (ICONC3)
Mody University of Science and Technology, Lakshmangarh, Feb 21-22, 2020

Robust H∞ Control Approach for Trajectory Tracking of Twin Rotor MIMO System

Ayush
Department of Electrical Engineering
Delhi Technological University
(Formely Delhi College of Engineering)
New Delhi, India
ayushaero@gmail.com

Hem Prabha
Department of Electrical Engineering
Delhi Technological University
(Formely Delhi College of Engineering)
New Delhi, India
prabhasingh1686@gmail.com

Rajul Kumar
Department of Electrical Engineering
Delhi Technological University
(Formely Delhi College of Engineering)
New Delhi, India
rajulkumar990@gmail.com

***Abstract*—The paper emphasis a simplified H∞ control strategy designed for trajectory tracking control for (TRMS) i.e. twin rotor multi-input multi-output system. Twin Rotor MIMO System (TRMS) is a benchmarked non-linear aero dynamical laboratory model having striking comparison like a helicopter but it doesn't fly. It is having 2-DOF (Degree of freedom), which corresponds to pitch and yaw angles determining the vertical and horizontal position of the TRMS. This paper elaborates a robust H∞ controller designed strategy for cross-coupled TRMS model as it guarantees robustness and good performance. The performance of the controller is tested and response of yaw and pitch are experimentally obtained through extensive simulation in MATLAB.**

Keywords—TRMS, 2-DOF, Mix sensitivity Problem, Loop shaping, Robust H∞ Control, Trajectory Tracking

I. INTRODUCTION

In past few years there has been a phenomenal interest in UAV[s] (Unmanned Aerial Vehicles). The Twin rotor MIMO system (TRMS) is a benchmarked aero dynamical laboratory experimental set-up i.e. a prototype of helicopter having striking similarity like it but doesn't fly. The laboratory set-up Twin Rotor MIMO system contains a beam which is pivoted on its base in this manner that it can move freely in both planes i.e. horizontally and vertically respectively. It has two rotors each at the end of the pivoted beam known as main or primary rotor and tail rotor respectively which are driven by 42-watt DC motors having operating voltage of 12V. TRMS is a highly non-linear system due to presence of cross coupling effect between its rotors and unstable dynamics [1]. Hence for the controlling of TRMS i.e. stabilization and tracking of the trajectory is a very daunting job and it has popularly attained a lot of research interests.

Robustness is an important aspect of control system design as the plant or systems are penetrable to the unpredictable external disturbances and noise. In recent years, H∞ controller has become a good alternative for controlling of a highly non-linear system having uncertainties [2],[3]. The conventional controller such as PID is easily realizable for TRMS but it hardly meets the demand of robustness because of presence of tuning of parameters i.e. k_p, k_i and k_d. So, the above drawbacks have been overcome to large extent by robust H∞ controller [4].

The aim of this paper is to explain the idea of H∞ optimal feedback controller design for TRMS. The paper briefly elaborates the loop shaping methodology to design H∞ feedback controller for trajectory tracking of Twin Rotor MIMO system. The simulation results of this paper illustrate that H∞ controller has better tracking performances along with lesser control efforts when compared to conventional controllers like PID.

This paper is organized into six sections, starting from Introduction followed by section II which described the modelling and dynamics of the TRMS. Section III illustrates the design methodology of H∞ controller. Section IV explains the synthesis of controller for TRMS model. In section V simulation results are presented with conclusion in section VI, followed by references.

II. TRMS DYNAMICS & MATHEMATICAL MODELING

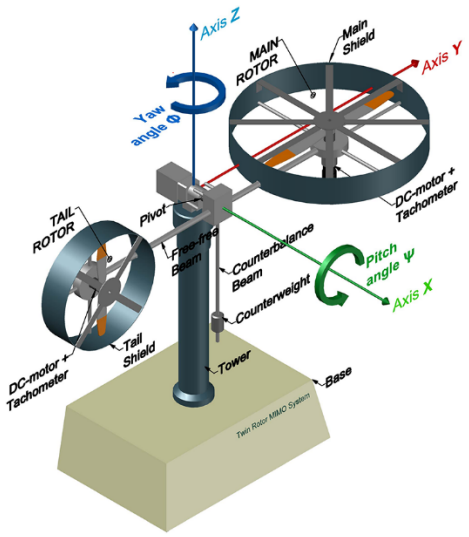


Fig. 1. Free Body diagram of TRMS

TRMS free body diagram is shown in Fig.1. It is an under actuated MIMO system which represents no less than one of the state i.e. (yaw angle φ and $\dot{\varphi}$ or pitch angle ψ and $\dot{\psi}$) in an argument of the nonlinear function[1],[5].

A. Dynamical Momentum Equations of TRMS

According to the free body diagram, the following TRMS dynamical model momentum mechanics for vertical and horizontal is derived as equations (1) and (2) respectively.

$$I_1\ddot{\psi} = a_1\tau_1^2 + b_1\tau_1 - M_g sin\psi - B_{1\psi}\dot{\psi} - B_{2\psi}sign(\dot{\psi}) - K_{gy}M_1\dot{\varphi}\cos(\psi) \quad (1)$$

$$I_2\ddot{\varphi} = a_2\tau_2^2 + b_2\tau_2 - B_{1\varphi}\dot{\psi} - B_{2\varphi}Sign(\dot{\varphi}) - \frac{k_c(T_o s+1)}{(T_p s+1)}\tau_1 \quad (2)$$

978-1-7281-1420-0/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 23:02:14 UTC from IEEE Xplore. Restrictions apply.

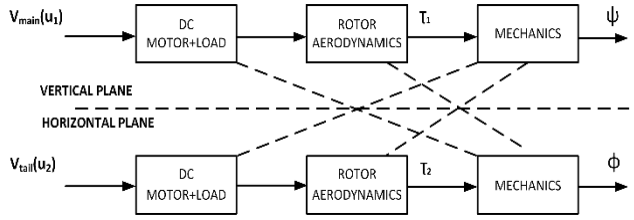


Fig. 2. Block diagram of TRMS coupled model

B. Actuator Momentum Equations

The actuator motor transfer functions for pitch and yaw are given in (3) and (4) respectively.

$$\tau_1 = \frac{K_1}{T_{11}s+T_{10}}u_1 \tag{3}$$

$$\tau_2 = \frac{K_2}{T_{21}s+T_{20}}u_2 \tag{4}$$

Table I gives the approximate model parameters of the TRMS.

TABLE I. TWIN ROTOR MIMO SYSTEM PARAMETERS AND VALUES

TRMS model Parameters	Values
$\boldsymbol{I_1}$ Moment of Inertia (Pitch Rotor)	$\boldsymbol{6.8\times10^{-2}Kgm^2}$
$\boldsymbol{I_2}$ Moment of Inertia (Yaw Rotor)	$\boldsymbol{2\times10^{-2}Kgm^2}$
$\boldsymbol{a_1}$ Pitch Rotor Static Constant	$\boldsymbol{0.0135}$
$\boldsymbol{b_1}$ Pitch Rotor Static Constant	$\boldsymbol{0.0924}$
$\boldsymbol{a_2}$ Yaw Rotor Static Constant	$\boldsymbol{0.02}$
$\boldsymbol{b_2}$ Yaw Rotor Static Constant	$\boldsymbol{0.09}$
$\boldsymbol{M_g}$ Gravitational Momentum	$\boldsymbol{0.32\ N-meter}$
$\boldsymbol{B_{1\psi}}$ Frictional Momentum (Pitch Rotor)	$\boldsymbol{6\times10^{-3}Nms/rad}$
$\boldsymbol{B_{2\psi}}$ Frictional Momentum (Pitch Rotor)	$\boldsymbol{1\times10^{-3}Nms^2/rad}$
$\boldsymbol{B_{1\varphi}}$ Friction Momentum (Yaw Rotor)	$\boldsymbol{1\times10^{-1}Nms/rad}$
$\boldsymbol{B_{2\varphi}}$ Friction momentum (Yaw Rotor)	$\boldsymbol{1\times10^{-2}Nms^2/rad}$
$\boldsymbol{K_{gy}}$ Gyroscopic Momentum	$\boldsymbol{0.05\ sec/rad}$
$\mathbf{k_1}$ Pitch DC Motor Gain	$\boldsymbol{1.1}$
$\boldsymbol{k_2}$ Yaw DC Motor Gain	$\boldsymbol{0.8}$
$\boldsymbol{T_{11}}$ Pitch Motor Moment Constant	$\boldsymbol{1.1}$
$\boldsymbol{T_{10}}$ Pitch Motor Moment Constant	$\boldsymbol{1}$
$\boldsymbol{T_{21}}$ Yaw Motor Moment Constant	$\boldsymbol{1}$
$\boldsymbol{T_{20}}$ Yaw Motor Moment Constant	$\boldsymbol{1}$
$\boldsymbol{T_p}$ Cross-coupling Constant	$\boldsymbol{2}$
$\boldsymbol{T_0}$ Cross-coupling Constant	$\boldsymbol{3.5}$
$\boldsymbol{k_c}$ Cross-coupling Gain	$\boldsymbol{-0.2}$

The linearized model in the state-space form of TRMS is written in (5) and (6) as

$$G_P = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \tag{5}$$

$$G_{p(s)} = \begin{bmatrix} \frac{1.246}{(s+0.833)(s^2+0.08824s+4.706)} & 0 \\ \frac{1.4823(s+0.2857)}{s(s+5)(s+0.833)(s+0.5)} & \frac{3.6}{s(s+7)(s+1)} \end{bmatrix} \tag{6}$$

III. DESIGN METHODOLOGY

In figure 2. considering the multivariable feedback control system. The motive is to design a controller having gain K(s) of the plant G(s) in such a manner that the closed-loop system attains specified performance with disturbances *d* and noise *n* present in it.[6]

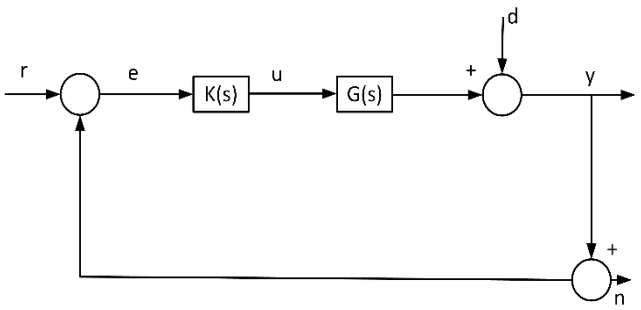


Fig. 3. Block Diagram representation of Multivariable Feedback System

For the illustration of the performances and multivariable stability margins for such type of systems, the singular values for the closed-loop transfer function are being used i.e. matrices from input r to each of the three outputs y, u, and e. By convention the matrices S(s) and T(s) are commonly known as the sensitivity and complementary sensitivity function of the plant and L(s) is known as nominal open loop transfer function are defined as follows:

$$L(s) = G(s)K(s) \tag{7}$$

$$S(s) = \left(I + L(s)\right)^{-1} \tag{8}$$

$$T(s) = L(s)/\left(I + L(s)\right)^{-1} \tag{9}$$

For obtaining good disturbance rejection and tracking of the reference signal in low frequency region, the output sensitivity should be kept low where the presence of disturbances is significant.

The singular value i.e. S(jω) determines the disturbance attenuation which can be written as

$$\bar{\sigma}\big(S(j\omega)\big) \ll 1 \tag{10}$$

$$\bar{\sigma}\big(S(j\omega)\big) \le \bar{\sigma}|W_S^{-1}| \qquad \forall\ \omega \tag{11}$$

where $\bar{\sigma}|W_S^{-1}|$ is known as disturbance attenuation factor.

The transfer function of the input r to control signal u is given by

$$\frac{u}{r} = Q(s) = \frac{K(s)}{1+G(s)K(s)} = K(s)S(s) \tag{12}$$

The most important factors at high frequencies are the robust stability and sensor noise rejection in the face of uncertainties present due to unmodeled dynamics and non-linearities. However, the control energy should be as low as possible and can be achieved by keeping the output control sensitivity low

$$\bar{\alpha}\big(K(j\omega)S(j\omega)\big) \ll 1 \tag{13}$$

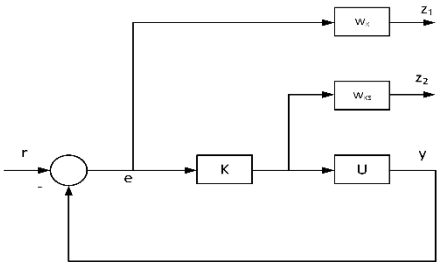


Fig. 4. Mixed Sensitivity Control Diagram

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 23:02:14 UTC from IEEE Xplore. Restrictions apply.

The weights $W_{ks}(s) = 1/K_{max}(s)S_{max}(s)$ gives the description of the shape of $K(s)S(s)$ which are satisfied by specifying upper limit of the singular value of $K(s)S(s)$ Therefore the robust stability and the sensor noise rejection are given as following in (14) and (15)

$$\bar{\sigma}\big(K(j\omega)S(j\omega)W_{ks}(j\omega)\big) < 1 \tag{14}$$

$$\Rightarrow \bar{\sigma}\big(K(j\omega)S(j\omega)\big) < \bar{\sigma}(W_{ks}^{-1}(j\omega)) \quad \forall\, \omega \tag{15}$$

The robust control problem can be referred as mixed loop shaping synthesis problem also known H_∞ control problem to get an internally stabilizing controller [8]

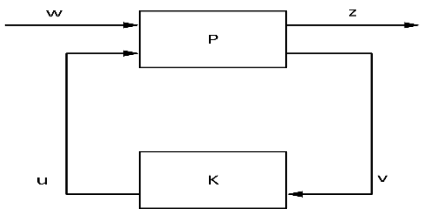


Fig. 5. Generalized plant and controller configuration

The input- output relationship of the plant shown in figure.5 can be written as (16)-(19)

$$u = K(s)v \tag{16}$$

$$\dot{x} = Ax + B_1 w + B_2 u \tag{17}$$

$$z = C_1 x + D_{11} w + D_{12} u \tag{18}$$

$$y = C_2 x + D_{21} w + D_{22} u \tag{19}$$

$$\begin{bmatrix} z \\ y \end{bmatrix} = P \begin{bmatrix} w \\ u \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} w \\ u \end{bmatrix} \tag{20}$$

The system which is to be controlled can be represented in matrix form (21) and (22)

$$P(s) = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix} \tag{21}$$

$$F_l(P,K) = P_{11} + P_{12}K(I - P_{22}K)^{-1}P_{21} \tag{22}$$

The $F_l(P,K)$ mapping is also known as linear fractional transformation. Now, the controller is to be designed so that the norm of H∞ i.e. the $F_l(P,K)$ is to be limited by a known constant gamma (γ) which reflects the desired level performances for closed-loop system. In the other words, controller gain K is to be found such that $F_l(P,K)$ and it is internally stable and satisfies the given condition in (23)

$$\|F_l(P,K)\|_\infty < \gamma \tag{23}$$

Using sufficient condition for robust performance can be written as (24)

$$\|F_l(P,K)\|_\infty = \|W_{ks}KS\|_\infty < \gamma \tag{24}$$

An internally stable closed loop map is generated by H∞ controller synthesis and hence it minimizes the H∞ cost function.

IV. H∞ CONTROLLER SYNTHESIS FOR TWIN ROTOR MIMO SYSTEM

The synthesis of H_∞ controller for TRMS can be achieved by using loop shaping technique[6],[7]. The main task is to design a controller (auto-pilot) for horizontal and vertical movement control so that the TRMS can track a predefined trajectory.

A. Selection of Weight

In low frequency ranges the sensitivity transfer function must be small for proper tracking and disturbances rejection. This may be achieved by determining the sensitivity function which is lower than the frequency dependent weight (W_S).

$$|S| \le |W_S^{-1}| \tag{25}$$

Whereas in high frequency ranges the complementary sensitivity transfer function should be as small as possible for rejection of noises which may be achieved by determining the complementary sensitivity lower than the frequency dependent weight (Wks).

$$|T| \le |W_{KS}^{-1}| \tag{26}$$

The term WS and Wks are our tuning parameters and generally it required few iterations for obtaining such weights that yields a appropriate controller.

$$W_S = \frac{S/M + \omega_O}{S + \omega_O A} \tag{27}$$

Here, A<1 specifies maximum steady state offset allowed whereas ω_O is bandwidth and M denotes the peak of the sensitivity function. For the desired controller synthesis, the upper limit of the required sensitivity loop is given by the inverse of WS whereas the inverse of Wks limits the control input u. The motive is to obtain controller gain K(s) that will minimize a weighted norm combining the above two functions as follows (28)

$$\left\| \begin{matrix} W_s(j\omega)s(j\omega) \\ W_{ks}(j\omega)s(j\omega) \end{matrix} \right\|_\infty \tag{28}$$

The gain of the controller for pitch angle control is given by

$$u_1 = K_1 * e_1 \tag{29}$$

Where

$$K_1 = \begin{bmatrix} -61.6959 & -114.6403 & -1.848e+03 & -8.1663e+06 & 0.3504 & 4 \\ -1.4662 & -5.1544 & -79.2818 & -3.4466e+05 & 0.0722 & 0 \\ -1.0715 & -2.1627 & -63.1779 & -2.7866e+05 & 0.0845 & 0 \\ -0.9759 & 17.6672 & -9.1659 & -8.5440e+04 & 37.2693 & 0 \\ 73.0648 & 136.4552 & 2.2035e+03 & 9.7431e+06 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Similarly, the controller gain for yaw angle control is given by

$$u_2 = K_2 * e_2 \tag{30}$$

$$K_2 = \begin{bmatrix} -1.9478e+03 & 1.0076e+03 & 3.2850e+07 & -0.4422 & 3 \\ 400.5213 & -209.6387 & -6.7140e+06 & 0.0193 & 0 \\ 17.3451 & 9.0453 & -5.9430e+05 & 16.6124 & 0 \\ -1.5222e+03 & 787.7518 & 2.5695e+07 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

V. RESULTS AND DISCUSSION

As explained in above sections, H∞ controller is designed and implemented on TRMS for trajectory tracking and controlling also robustness analysis and performance of the TRMS is studied through simulation in MatLab. The

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 23:02:14 UTC from IEEE Xplore. Restrictions apply.

responses of TRMS for pitch and yaw angle control is depicted as shown in figures (6),(7) and (8). A unit step input is given to the system for tracking of the desired trajectory. The output of controller is taken out to analyze the robustness of H∞ multivariable feedback control of TRMS. The condition of inequalities is plotted for verification of robustness. It clearly shows that the H∞ controller for TRMS is stable even in the presence of disturbances.

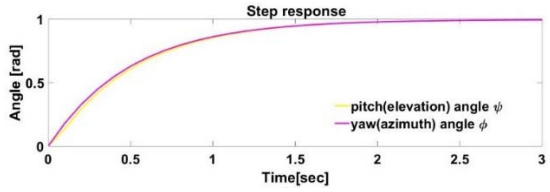


Fig. 6. Unity step input response of H_∞ controller for TRMS

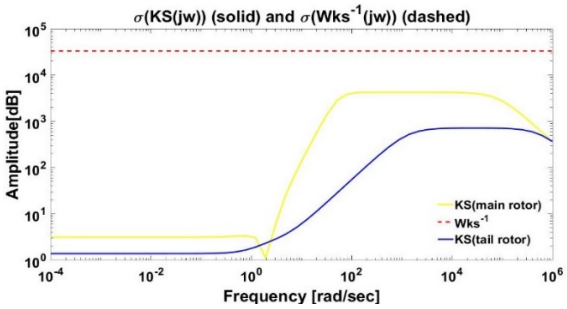


Fig. 7. Robust stability condition for TRMS

VI. CONCLUSION

This paper presents a simplified H_∞ control strategy for Twin rotor MIMO system by loop shaping control mix sensitivity method. The paper first addresses the dynamics and mathematical modelling of TRMS and then design methodology of H_∞ mixed sensitivity controllers. From the results obtained it is concluded that with the help of good selection of the weight of sensitivity function, the H_∞ controller can be the excellent for stabilization of Twin Rotor MIMO System.

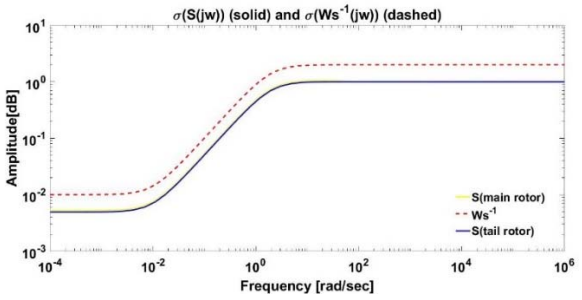


Fig. 8. Sensor Noise Rejection for TRMS

REFERENCES

[1] Feedback Instruments, "Twin Rotor MIMO System," 2016.

[2] N. Almtireen,H. Elmoaqet, and M. Ryalat, "Linearized Modelling and Control for a Twin Rotor System," *Automatic Control and Computer Sciences*, vol. 52, no. 6, pp. 539–551, 2019.

[3] P. Wen and Y. Li, "Twin rotor system modeling, de-coupling and optimal control," in *IEEE International Conference on Mechatronics and Automation*, 2011, pp. 1839–1842.

[4] T.K Liu and Juang: "A Single Neuron PID Control for Twin Rotor MIMO System", IEEE/ASME International Conference on Advanced Intelligent Mechatronics, 2009.

[5] Chalupa P, Pikryl J, Novk J. Modelling of twin rotor MIMO system. Procedia Eng.2015; 100:249-258 25th DAAAM International Symposiums on Intelligent Manufacturing and Automation, 2014.

[6] Pratap B, Purwar S (2010) Neural network observer for twin rotor MIMO system: an LMI based approach. In: Proceedings of the international conference on modelling, identification and control, Okayama, Japan.

[7] Te-wei Lu, Peng Wen. "Time Optimal and Robust Control of Twin Roor System", 2007 IEEE International Conference on Control and Automation.

[8] Q. Ahmed, A, I. Bhatti, S. Iqbal "Nonlinear Robust Decoupling controller design for Twin Rotor System", Proc. Of 7th Asian Control Conference Hong Kong, China 2009.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 23:02:14 UTC from IEEE Xplore. Restrictions apply.